

Efficient Attributed Scatter Center Extraction Based on Image-Domain Sparse Representation

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Abstract—As an efficient way to interpret the measurements of high-frequency synthetic aperture radar (SAR), an attributed scattering center (ASC) model provides concise and physically relevant features of complex targets. However, accurate extractions of ASCs have been heavily penalized by high memory requirements and computational complexity. We propose to convert SAR measurements to sparse representations in the image domain where the ASC model parameters can be estimated by using an orthogonal matching pursuit (OMP) algorithm or its Newtonized variation. Two important new properties of the ASC model are unveiled in the image domain, namely, “translatability” and “additivity.” The properties can help save the dictionary of OMP from sampling the position and length parameters. The atoms of the dictionary become localized, thereby reducing the dictionary size and accelerating ASC extractions. Extensive experiments are conducted based on open-source XPATCH Backhoe data, measured MSTAR data, and synthetic backscatter data. The results show that the proposed approach is able to outperform existing image-domain algorithms in terms of accuracy and noise resistance, and outperform existing frequency-domain algorithms in terms of memory requirement and runtime.

Index Terms—Synthetic aperture radar (SAR) image, attributed scattering model (ASM), orthogonal matching pursuit (OMP), quasi-Newton algorithm.

I. INTRODUCTION

TYPICAL synthetic aperture radar (SAR) image-based automatic target recognition (ATR) techniques can be classified into two categories, according to the features that the algorithms exploit: data-driven algorithms and physical-model-based algorithms [1]. Data-driven algorithms extract imagery features and train a classifier with a large dataset. The classifier can be a support vector machine (SVM) [2], Bayesian classifier [3], and convolutional neural network (CNN) [4]. The algorithms are relatively easy to deploy, but difficult to generalize under all operating conditions (e.g., the configurations

of a SAR system and the orientation of the target) due to the fact that the imagery features are susceptible to operating conditions. Physical-model-based algorithms use physical features based on the physical characteristics of the target and the radar target signatures. The physical features contain explicit physical attributes (typically, the scattering centers, size, and the shape of a target) and are independent of operating conditions. Physical-model-based algorithms extract physical features from the tested SAR image chip of a target, and classify the target by comparing the features with templates constructed from a computer-aided design (CAD) model or features from pre-stored SAR image chips of different targets [5], [6]. Parametric scattering models are the representative type of physical models, which build up the relations between radar backscatter and the physical features of a target (such as the scattering centers of the target, as well as the shape, size and the reflectiveness of each scatter).

From the geometrical theory of diffraction, a high-frequency scattering response from a complex object can be well approximated by a superposition of responses from individual scatters [7]. Parametric models of isolated radar scattering responses, developed from the theory, play an important role in scattering mechanism analysis and feature-based target discrimination and identification. Significant progress has been made developing parsimonious physics-based parametric models [8]–[12]. Gerry *et al.* [11], [12] developed a new parametric attributed scattering model (ASM), which characterizes the dependence of the response on the frequency and aspect angle for both localized and distributed scattering mechanisms. The parameter set of the model provides a physical description of the scatters and, in turn, rich information on object geometries. The parametric scattering models have been successfully applied to SAR ATR [10], [13]–[23].

Attributed scattering center (ASC) parameter estimation is a high-dimensional, nonlinear, non-convex problem. Two types of the solutions to the problem have been developed, namely, image-domain algorithms and frequency-domain algorithms. Existing image-domain algorithms transform radar backscatters into the image domain and extract ASCs based on the segmentation of obtained images [12], [14], [24]–[26]. The state-of-the-art image-domain algorithm was presented in [26] with several variations deployed in [22], [27]. The key idea of [26] is first to segment an image and then carry out parameter estimations, e.g., by using the maximum-likelihood estimation on each of the segments. However, the algorithm may undergo an inaccuracy resulting from inadequate image segmentation,

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especially under low signal-to-noise ratios (SNRs) [28], as will be revealed experimentally in this paper. The reason is that SAR images are intricate, and the segmentation may not be precise enough to separate each ASC correctly.

Frequency-domain algorithms can avoid image segmentation preprocessing. Duan *et al.* [29] utilized a relaxation algorithm and performed feature extraction directly in the frequency domain. Min *et al.* [30] estimated the ASC model parameters in the frequency domain by using an orthogonal matching pursuit (OMP) algorithm. In [31], OMP was combined with the relaxation algorithm to estimate the ASC model parameters in the frequency domain. However, the frequency-domain algorithms typically suffer from a high computational complexity and storage requirement.

Li *et al.* [32] and Cong *et al.* [28] combined the advantages of image-domain and frequency-domain algorithms. In [32], ASCs were first coarsely extracted in the image domain, and the ASC parameters were precisely estimated based on incremental sparse Bayesian learning in the frequency domain. Cong *et al.* [28] developed an ASC extraction algorithm based on Levy random field in a nonparametric Bayesian framework, and initialized the estimation parameters according to the SAR image. Both [28] and [32] were under the assumption that the SAR image could provide reliable information. They would fail if the SAR image is intricate, and the image segmentation algorithms are hard to be effective. The algorithms would also suffer from high computational complexities.

This paper proposes a new ASC extraction approach, which converts SAR signals into the image domain and estimates the ASC parameters efficiently based on the sparse representation of the SAR images. A key contribution is a new Newtonlized OMP (NOMP)-based ASC parameter estimation algorithm, which pre-generates a dictionary of multidimensional elements (referred to as “atoms”) underlying the image of every possible ASC. By looking up the dictionary for the most correlated atoms with a SAR image, meaningful ASCs are extracted one after another from the image. Another critical contribution is that we unveil two important properties of the ASC models in the image domain, namely, “*translatability*” and “*additivity*.” The two properties are exploited to reduce the dimension of the dictionary atoms by avoiding sampling the position and length parameters of the ASCs in the proposed NOMP-based algorithm. The atoms become localized. Hence, the dictionary size is reduced, and the dictionary lookup of the algorithm is substantially sped up. Extensive experiments are conducted to verify the proposed approach, where open-source XPATCH Backhoe data, measured MSTAR data, and synthetic backscatter data are tested. It is shown that the proposed approach outperforms the state-of-the-art image-domain algorithm [26] in terms of accuracy and noise resistance, and outperforms the state-of-the-art frequency-domain algorithm [32] in terms of memory requirement and runtime.

The rest of the paper is arranged as follows. In Section II, we provide an overview of the ASC model. In Section III, we formulate the problem of interest and develop the new NOMP-based framework to solve the problem in the image domain. In

Section IV, we prove the *translatability* and *additivity* of ASCs, with which both the memory requirement and computational complexity of NOMP are reduced substantially. Experimental evaluations of the proposed technique are provided in Section V, followed by conclusions in Section VI.

Notation: Superscripts $(\cdot)^T$, $(\cdot)^H$ and $(\cdot)^*$ denote transpose, Hermitian transpose, and conjugate, respectively. $(\cdot)^\dagger$ denotes pseudo-inverse. $\text{tr}(\cdot)$ stands for trace. $\|\cdot\|_2$ denotes l_2 norm. $\|\cdot\|_F$ denotes Frobenius norm. $\mathbf{O}$ stands for a zero matrix. Vector $(\sim, \vartheta_i, \sim)$ indicates that the entries remain unchanged in the vector, except ϑ_i . $a := b$ assigns the value of b to a .

II. THE ASC MODEL

Based on high-frequency approximations [7], a radar backscatter can be approximated by a combined effect of the backscatters of different ASCs. Different from the widely used point scattering (PS) model which characterizes scattering centers with two position parameters and one backscattered coefficient parameter, the ASC model depicts every scattering center with seven parameters and provides a concise and physically meaningful description of a target radar signature.

The ASC model can be expressed as [33]

$$\tilde{S}(f, \varphi; \Theta) = \sum_{q=1}^Q \tilde{E}_q(f, \varphi; \theta_q) + n(f, \varphi), \quad (1)$$

where $(\tilde{\cdot})$ and $(\cdot)$ are a pair of transforms in the frequency and image domains, respectively; Θ is the set of all parameters; $\tilde{S}(f, \varphi; \Theta)$ denotes the radar backscatter and Θ is suppressed in the rest of this paper for brevity; f is the frequency; φ is the aspect angle; Q is the number of ASCs; and $n(f, \varphi)$ is the additive complex white Gaussian noise.

Here, $\theta_q = (A_q, \alpha_q, L_q, \bar{\varphi}_q, \gamma_q, x_q, y_q)$ collects all the model parameters of the q -th ASC to be estimated. A_q is the backscattering coefficient. α_q is the frequency dependence. x_q and y_q are the range and cross-range locations, respectively. L_q , $\bar{\varphi}_q$ and γ_q provide the aspect dependence of scattering.

In (1), $\tilde{E}_q(f, \varphi; \theta_q)$ is the response of the q -th ASC, and can be written as

$$\begin{aligned} \tilde{E}_q(f, \varphi; \theta_q) = & A_q \left(j \frac{f}{f_0} \right)^{\alpha_q} \text{sinc} \left(\frac{2\pi f}{c} L_q \sin(\varphi - \bar{\varphi}_q) \right) \\ & \cdot \exp \{ -2\pi f \gamma_q \sin \varphi \} \\ & \cdot \exp \left\{ -j \frac{4\pi f}{c} (x_q \cos \varphi + y_q \sin \varphi) \right\}, \quad (2) \end{aligned}$$

where f_0 is the radar center frequency, and c is the light speed.

For localized ASCs (e.g., trihedral, top hat, sphere, and corner diffraction), $L_q = \bar{\varphi}_q = 0$, and γ_q measures the small aspect dependence. For distributed ASCs (e.g., dihedral, cylinder, edge broadside and edge diffraction), $\gamma_q = 0$, L_q is the length of the ASC, and $\bar{\varphi}_q$ is its orientation angle [28].

The frequency dependence α and the length L differ among the scattering geometries. L indicates whether the ASC is localized or distributed, while α depends on the curvature of the

TABLE I
PHYSICAL INTERPRETATION OF PARAMETERS α AND L [28]

Scattering Geometry	Iconic Representation	α	L
Dihedral		1	$L > 0$
Cylinder		0.5	$L > 0$
Edte Broadside (EB)		0	$L > 0$
Edge Diffraction (ED)		-0.5	$L > 0$
Trihedral		1	$L = 0$
Top Hat		0.5	$L = 0$
Sphere		0	$L = 0$
Corner Diffraction (CD)		-1	$L = 0$

scattering primitive [34], [35]. According to α and L , the ASCs can be classified into eight canonical scattering geometries, as listed in Table I [28].

III. PROPOSED NOMP-BASED ASC EXTRACTION FRAMEWORK

In this paper, the problem of interest is to estimate the ASC model parameters, namely, the number of ASCs that represent a target, i.e., Q , and the parameters of each ASC, i.e., θ_m , $m = 1, 2, \dots, Q$. The problem can be written as

$$\min Q, s.t. \left\| \tilde{S}(f_k, \varphi_h) - \sum_{q=1}^Q \sigma_q \tilde{E}(f_k, \varphi_h; \theta_q) \right\|_2 \leq \varepsilon, \quad (3)$$

where f_k and φ_h are the k -th and h -th samples of the discretized versions of f and φ , respectively. $\tilde{S}(f_k, \varphi_h)$ is the measured data in the frequency domain. $\tilde{E}(f_k, \varphi_h; \theta_q)$ is the signal of the q -th ASC; $\varepsilon > 0$ is the q -th ASC of the target. $\tilde{S}(f_k, \varphi_h)$ and $\tilde{E}(f_k, \varphi_h; \theta_q)$ are the discretized versions of $\tilde{S}(f, \varphi)$ and $\tilde{E}(f, \varphi; \theta_q)$, respectively. The frequency-domain signals, $\tilde{S}(f_k, \varphi_h)$, are obtained by first applying the Hilbert transform to the time-domain received signals and down-converting to the baseband, then correlating the baseband analytic signals with the transmitted signals (i.e., pulse compression), followed by the Fourier transform to finally obtain the frequency-domain data [36].

Problem (3) is a sparse representation problem due to the fact that man-made targets are usually dominated by echoes from a small number of ASCs. We propose to transform Problem (3) to the image domain where we reveal two important properties to facilitate solving (3), as will be described in Section IV. To achieve this, we first introduce a 2D SAR imaging operator which transforms a discrete phase history $\tilde{S}(f_k, \varphi_h)$ to a continuous SAR image [33]

$$\mathcal{B}_d \{ \cdot \} = \frac{1}{2\pi} \sum_{k=1}^K \sum_{h=1}^H \{ \cdot \} e^{j \frac{4\pi f_k}{c} (x \cos \varphi_h + y \sin \varphi_h)}. \quad (4)$$

In essence, this operator is a non-uniform discrete inverse Fourier transform operator. By applying the operator to $\tilde{S}(f_k, \varphi_h)$ and $\tilde{E}(f_k, \varphi_h; \theta_q)$, Problem (3) can be equivalently

transformed from the discrete frequency domain to the continuous image domain, as given by

$$\min Q, s.t. \left\| S(x, y) - \sum_{q=1}^Q \sigma_q D(x, y; \theta_q) \right\|_2 \leq \varepsilon, \quad (5)$$

where $S(x, y)$ and $D(x, y)$ are the continuous images transformed from $\tilde{S}(f_k, \varphi_h)$ and $\tilde{E}(f_k, \varphi_h; \theta_q)$, respectively.

A state-of-the-art solver for sparse representation problems is the OMP algorithm [37]. OMP is a sparse approximation algorithm that finds the “best matching” projections of multidimensional data into the span of an over-complete (i.e., redundant) dictionary. The basic idea is to approximately represent a signal from the Hilbert space as a weighted sum of a finite number of functions (i.e., atoms) taken from the dictionary. [38] generalizes the OMP to the continuum by using Newton refinements. Hence, the resulting algorithm is referred to as NOMP.

A. Dictionary Construction

Like OMP, NOMP needs to construct a dictionary Φ :

$$\Phi = \left\{ \hat{D}(x, y; \theta_q) \mid \theta_q \in \Theta \right\}; \quad (6)$$

$$\Theta = \Theta_{loc} \cup \Theta_{dis};$$

$$\Theta_{loc} = \Theta_A \times \Theta_{\alpha_{loc}} \times \Theta_{L_{loc}} \times \Theta_{\bar{\varphi}_{loc}} \times \Theta_{\gamma_{loc}} \times \Theta_x \times \Theta_y;$$

$$\Theta_{dis} = \Theta_A \times \Theta_{\alpha_{dis}} \times \Theta_{L_{dis}} \times \Theta_{\bar{\varphi}_{dis}} \times \Theta_{\gamma_{dis}} \times \Theta_x \times \Theta_y, \quad (7)$$

where $\hat{D}(x, y; \theta_q) = D(x, y; \theta_q) / \|D(x, y; \theta_q)\|_2$ is the normalized ASC image; Θ_{loc} and Θ_{dis} are the sets of the parameter tuples for localized ASCs and distributed ASCs in the dictionary, respectively. Θ_A , Θ_x , Θ_y , Θ_L , and Θ_γ are the sets of parameters A , α , x , y , L , $\bar{\varphi}$, and γ , respectively; $\Theta_{\alpha_{loc}}$, $\Theta_{L_{loc}}$ and $\Theta_{\bar{\varphi}_{loc}}$ are the sets of parameters α , L , and $\bar{\varphi}$ of localized ASCs in the dictionary, respectively; $\Theta_{\alpha_{dis}}$, $\Theta_{L_{dis}}$ and $\Theta_{\bar{\varphi}_{dis}}$ are the sets of parameters α , L , and $\bar{\varphi}$ of distributed ASCs in the dictionary, respectively; and $\times$ stands for the Cartesian product.

According to the definition of the ASC model, we define $\Theta_{\gamma_{dis}} = \Theta_{L_{loc}} = \Theta_{\varphi_{loc}} = \{0\}$. $\Theta_A = \{1\}$, since the atoms are normalized in terms of amplitude. $\Theta_{\alpha_{loc}} = \Theta_{\alpha_{dis}} = \Theta_{\gamma_{loc}} = \{0\}$ due to the fact that the parameters α and γ contribute negligibly to the ASC signals [29]. $\Theta_{L_{dis}} = \{2\Delta L, 4\Delta L, \dots, 2N_L \Delta L\}$, $\Theta_{\bar{\varphi}_{dis}} = \{\bar{\varphi}_1, \bar{\varphi}_2, \dots, \bar{\varphi}_{N_{\bar{\varphi}}}\}$, $\Theta_x = \{x_1, x_2, \dots, x_{N_x}\}$, and $\Theta_y = \{y_1, y_2, \dots, y_{N_y}\}$ collect the parameters uniformly sampled from L , φ , x and y , respectively.

B. NOMP-Based ASC Extraction

Apart from the construction of the dictionary Φ , NOMP consists of four iterative steps, namely, atom selection, atom refinement, projection, and residue evaluation. As a result, a signal can be sparsely approximated by a set of refined atoms. When applied to the image-domain ASC extraction, the four steps operate as follows.

Step 1: Given the dictionary Φ and the residual image $R(x, y)$ after canceling $(i - 1)$ estimated ASCs from $S(x, y)$, NOMP estimates the parameter of the i -th ASC, denoted by θ_i , by selecting the most matched atom $\hat{D}(x, y; \theta_q) \in \Phi$ for $R(x, y)$. This can be done by evaluating the inner product of the SAR image and every atom in Φ , as given by

$$\theta_i = \arg \max_{\theta_q} \left| \iint R(x, y) \hat{D}^*(x, y; \theta_q) dx dy \right|, \quad (8)$$

Step 2: θ_i may not be optimal, as the ASC parameters contain continuous values. This step improves the estimation of the continuous parameters of the i -th ASC (i.e., $x, y, L, \bar{\varphi}$ and γ) by taking θ_i as the starting point and running Newton's method [38]:

$$\theta_{i,opt} = \arg \max_{\theta \in \{A\}} \underbrace{\left| \iint R(x, y) \hat{D}^*(x, y; \theta) dx dy \right|^2}_{J(\theta)}. \quad (9)$$

With reference to [39], the gradient of $J(\theta)$ with respect to θ is provided in (10), shown at the bottom of this page, where ∂_{ϑ_j} stands for $\frac{\partial}{\partial \vartheta_j}$, ϑ_j is the j -th entry of θ , and $\text{Re}(\cdot)$ stands for the real part of a complex value. Additionally, α can only take a small number of discrete values, and α_{opt} can be obtained by brute-force.

Step 3: A least-square estimation is carried out to evaluate the coefficients of the i selected atoms, as given by

$$\sigma^{i,opt} = \arg \min_{\sigma^i} \left\| S(x, y) - \sum_{q=1}^i \sigma_q^i D(x, y; \theta_{q,opt}) \right\|_2^2, \quad (11)$$

where $\sigma^i = (\sigma_1^i, \sigma_2^i, \dots, \sigma_i^i)$ collects the coefficients associated with the i atoms selected so far, and $\sigma^{i,opt} = (\sigma_1^{i,opt}, \sigma_2^{i,opt}, \dots, \sigma_i^{i,opt})$ collects the optimal coefficients which minimize the difference between the original image $S(x, y)$ and the reconstructed based on the i selected atoms.

Step 4: We update the residual image by canceling the i selected atoms, i.e., $R(x, y) := S(x, y) - \sum_{q=1}^i \sigma_q^{i,opt} D(x, y; \theta_{q,opt})$. Then, we estimate the next most significant ASC in $R(x, y)$ by repeating Steps 1–4 until a stop criterion is met.

The stop criterion is either a predefined threshold of the ratio of the residual signal to the original received signal, denoted by ε_R , i.e., $\|R(x, y)\|_2^2 / \|S(x, y)\|_2^2 < \varepsilon_R$, or an empirically defined maximum number of iterations, whichever is first reached. The maximum number of iterations depends on how complex a target is and is preferably large enough to allow the proposed

Algorithm 1: The Proposed NOMP-Based ASC Model Parameter Extraction Framework.

Input: $S(x, y), \Phi$

Output: $(\theta_{q,opt}, \sigma_q^{i,opt}), q = 1, 2, \dots, Q$

Initialization: $R(x, y) := S(x, y), \Phi_{Gen} = \emptyset, i := 1$

- 1: **while** The stop criterion is not met. **do**
 - 2: Coarsely estimate the i -th ASC by selecting from Φ the atom matching most $R(x, y)$, i.e., $D(x, y; \theta_i)$.
 - 3: Refine the estimation of the i -th ASC by taking Newton's method, and obtain $\theta_{i,opt}$.
 $\Phi_{Gen} := \Phi_{Gen} \cup \{D(x, y; \theta_{i,opt})\}$.
 - 4: Estimate the coefficients $\sigma^{i,opt}$ associated with $\theta_{i,opt}$ by using the least-square estimation.
 - 5: Update $R(x, y)$ by canceling the estimated atoms.
 - 6: $i := i + 1$.
 - 7: **end while**
-

algorithm to extract meaningful ASCs. For example, we set the number empirically to be 8 for simulated data and 30 for the Backhoe data in Section VI.

Algorithm 1 summarizes the proposed NOMP-based ASC model parameter extraction framework, where Φ_{Gen} is a set defined to collect the selected atoms.

C. Memory and Computational Overhead

In practice, a SAR image needs to be discretized to be a matrix. The proposed framework (and the dictionary reduction and acceleration techniques to be proposed in Section IV) can be readily adapted to discretized SAR images. This is because two-dimensional continuous functions, such as the residual image $R(x, y)$ and the atoms $\hat{D}(x, y; \theta_q)$, can be discretized to matrices $\mathbf{R}$ and $\hat{\mathbf{D}}(\theta)$. All the operations involved in the proposed technique, i.e., integration, convolution and inner product, can be readily implemented in the discrete image domain. For example, $\iint R(x, y) \hat{D}^*(x, y; \theta_q) dx dy$ in (8) can be written as $\text{tr}(\hat{\mathbf{D}}(\theta)^H \mathbf{R})$ in the discrete image domain. Integrations in the continuous image domain can be replaced by summations in the discrete image domain.

After discretization, the dictionary Φ can be large. There are $(N_L N_{\bar{\varphi}} + 1) N_x N_y$ atoms in Φ , where $N_L, N_{\bar{\varphi}}, N_x$, and N_y are the numbers of discrete levels for $L, \bar{\varphi}, x$, and y , respectively. Every atom is of the same size as the SAR image $S(x, y)$. For example, $N_L = N_{\bar{\varphi}} = 16$ and $N_x = N_y = 512$ for a typical 512×512 SAR image. The total memory would be as large

$$\begin{aligned} \partial_{\vartheta_j} J(\theta) &= 2 \text{Re} \left\{ \iint R^*(x, y) \partial_{\vartheta_j} \frac{\mathcal{B}_d(\tilde{S}(f_k, \varphi_h; \theta))}{\|\mathcal{B}_d(\tilde{S}(f_k, \varphi_h; \theta))\|_2} dx dy \cdot \left[\iint R^*(x, y) \frac{\mathcal{B}_d(\tilde{S}(f_k, \varphi_h; \theta))}{\|\mathcal{B}_d(\tilde{S}(f_k, \varphi_h; \theta))\|_2} dx dy \right]^* \right\} \\ \partial_{\vartheta_j} \frac{\mathcal{B}_d(\tilde{S}(f_k, \varphi_h; \theta))}{\|\mathcal{B}_d(\tilde{S}(f_k, \varphi_h; \theta))\|_2} &= \frac{\mathcal{B}_d(\partial_{\vartheta_j} \tilde{S}) \|\mathcal{B}_d(\tilde{S})\|_2^2 - \text{Re} \left[\iint \mathcal{B}_d(\partial_{\vartheta_j} \tilde{S})^* \mathcal{B}_d(\tilde{S}) dx dy \right] \mathcal{B}_d(\tilde{S})}{\|\mathcal{B}_d(\tilde{S})\|_2^3} \end{aligned} \quad (10)$$

as 32 TB. This poses serious requirements for memory and storage.

NOMP can be computationally demanding. For an $M \times N$ discretized SAR image, the atom selection (i.e., Step 1) involves $MN(N_L N_{\bar{\varphi}} + 1)N_x N_y$ multiplications. Hence, it has a computational complexity of $O(MN N_L N_{\bar{\varphi}})$. The complexity of the Newtonized refinement of the estimation (i.e., Step 2) is dominated by the imaging operator $\mathcal{B}_d(\cdot)$ in (10), which can be implemented by the celebrated back projection (BP) SAR imaging algorithm [40]. The BP algorithm has a complexity of $O(M^2 N \log(N))$. Consider an accuracy requirement of Newton's method ϵ . The overall complexity of the Newtonized refinement is $O(M^2 N \log(N) \log(\frac{1}{\epsilon}))$. The complexity of these two steps does not grow with the number of iterations (or, in other words, the number of meaningful ASCs of the target).

The least-square estimation in Step 3 involves a pseudo-inversion of an $i \times MN$ matrix, as given by

$$\sigma^{i,opt} = [\text{vec}(\mathbf{D}(\theta_1)), \text{vec}(\mathbf{D}(\theta_2)), \dots, \text{vec}(\mathbf{D}(\theta_i))]^\dagger \mathbf{S}, \quad (12)$$

where $\mathbf{S}$ and $\mathbf{D}$ are two $M \times N$ -dimensional matrices discretized from $S(x, y)$ and $D(x, y)$, respectively; and $\text{vec}(\cdot)$ stands for vectorization. The computational complexity of Step 3 is $O(i^2 MN)$. With the increasing iterations, Step 3 can increasingly dominate the computational complexity of Algorithm 1. The complexity can grow quadratically with iterations.

IV. DICTIONARY REDUCTION AND EXTRACTION ACCELERATION

A. Two Properties of ASC Models in Image Domain

Based on the operator (4), two properties of the ASC model are revealed in the image domain. The properties lay the cornerstone to reducing the memory requirement and computational complexity of the proposed NOMP-based ASC extraction framework.

Property 1 ("Translatability"): If two ASCs have the same parameters except for the two position parameters x and y , they have the same wave shape in the image domain.

Proof: Property 1 can be proved directly with the following:

$$\begin{aligned} D(x, y; \theta_0) &= \mathcal{B}_d \left\{ \tilde{E}(f_k, \varphi_h; \theta_0) \right\} \\ D(x, y; \theta'_0) &= \mathcal{B}_d \left\{ \tilde{E}(f_k, \varphi_h; \theta'_0) \right\} \\ &= D(x - \Delta x_0, y - \Delta y_0; \theta_0), \end{aligned} \quad (13)$$

where $D(x, y; \theta_0)$ is the SAR image corresponding to the phase history data $\tilde{E}(f, \varphi; \theta_0)$; $D(x, y; \theta'_0)$ is the SAR image corresponding to the phase history data $\tilde{E}(f, \varphi; \theta'_0)$; and

$$\begin{aligned} \theta_0 &= (A, \alpha, L, \bar{\varphi}, \gamma, x_0, y_0); \\ \theta'_0 &= (A, \alpha, L, \bar{\varphi}, \gamma, x_0 - \Delta x_0, y_0 - \Delta y_0). \end{aligned} \quad (14)$$

It is revealed in (13) that the SAR image formed from $\tilde{E}(f_k, \varphi_h; \theta'_0)$ can be obtained by linearly shifting the SAR image formed from $\tilde{E}(f_k, \varphi_h; \theta_0)$. Property 1 is proved. ■

Property 2 ("Additivity"): An ASC with length L is a linear combination of two ASCs with lengths L_1 and L_2 , where $L = L_1 + L_2$.

Proof: A sinc function can be seen as a linear combination of two sinc functions, as (15), shown at the bottom of this page. Then, we have

$$E(f, \varphi; \theta_0) = \frac{L_1}{L} E(f, \varphi; \theta_1) + \frac{L_2}{L} E(f, \varphi; \theta_2), \quad (16)$$

where

$$\begin{aligned} \theta_1 &= (A, \alpha, L_1, \bar{\varphi}, \gamma, x_0 - \frac{L_1 - L}{2} \sin \bar{\theta}, y_0 + \frac{L_1 - L}{2} \cos \bar{\theta}); \\ \theta_2 &= (A, \alpha, L_2, \bar{\varphi}, \gamma, x_0 + \frac{L - L_2}{2} \sin \bar{\theta}, y_0 - \frac{L - L_2}{2} \cos \bar{\theta}). \end{aligned} \quad (17)$$

As a result, Property 2 holds in the frequency domain.

By applying the imaging operator (4) to both sides of (16), we have the image-domain representation of Property 2 as follows.

$$S(x, y; \theta_0) = \frac{L_1}{L} S(x, y; \theta_1) + \frac{L_2}{L} S(x, y; \theta_2). \quad (18)$$

As a result, Property 2 is proved in the image domain. ■

Usually, the measured phase history of the SAR image is not in the baseband. We down-convert the frequency spectrum of the SAR image to the baseband by using the following SAR

$$\begin{aligned} \text{sinc} \left[\frac{2\pi f_k}{c} L \sin(\varphi_h - \bar{\varphi}) \right] &= \frac{1}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} \exp \left[-j \frac{4\pi f_k}{c} l \sin(\varphi_h - \bar{\varphi}) \right] dl \\ &= \frac{1}{L} \int_{-\frac{L}{2}}^{-\frac{L}{2} + L_1} \exp \left[-j \frac{4\pi f_k}{c} l \sin(\varphi_h - \bar{\varphi}) \right] dl + \frac{1}{L} \int_{\frac{L}{2} - L_2}^{\frac{L}{2}} \exp \left[-j \frac{4\pi f_k}{c} l \sin(\varphi_h - \bar{\varphi}) \right] dl \\ &= \frac{L_1}{L} \text{Sinc} \left[j \frac{2\pi f_k}{c} L_1 \sin(\varphi_h - \bar{\varphi}) \right] \cdot \exp \left\{ -j \frac{4\pi f_k}{c} \left[\left(x - \frac{L_1 - L}{2} \sin \bar{\varphi} \right) \cos \varphi_h + \left(y + \frac{L_1 - L}{2} \cos \bar{\varphi} \right) \sin \varphi_h \right] \right\} \\ &\quad + \frac{L_2}{L} \text{Sinc} \left[j \frac{2\pi f_k}{c} L_2 \sin(\varphi_h - \bar{\varphi}) \right] \cdot \exp \left\{ -j \frac{4\pi f_k}{c} \left[\left(x - \frac{L - L_2}{2} \cos \bar{\varphi} \right) \cos \varphi_h + \left(y + \frac{L - L_2}{2} \cos \bar{\varphi} \right) \sin \varphi_h \right] \right\} \end{aligned} \quad (15)$$

imaging operator:

$$\mathcal{B}\{\cdot\} = \frac{1}{2\pi} \sum_{k=1}^K \sum_{h=1}^H \{\cdot\} e^{j\frac{4\pi f_k}{c}(x \cos \varphi_h + y \sin \varphi_h)} e^{-j\frac{4\pi f_0}{c}x}. \quad (19)$$

With (19), the frequency-domain representation of a SAR image is in the baseband. The use of (19) can guarantee that the properties derived withstand under different pixel sizes. Specifically, (13) can be rewritten as

$$\begin{aligned} S(x, y; \theta_0) &= \mathcal{B}\{E(f_k, \varphi_h; \theta_0)\}; \\ S(x, y; \theta'_0) &= \mathcal{B}\{E(f_k, \varphi_h; \theta'_0)\} \\ &= S(x - \Delta x_0, y - \Delta y_0; \theta_0) e^{-j\frac{4\pi f_0}{c}\Delta x_0}. \end{aligned} \quad (20)$$

Therefore, (18) can be rewritten as

$$\begin{aligned} S(x, y; \theta_0) &= \frac{L_1}{L} S(x, y; \theta_1) \exp\left\{-j\frac{4\pi f_0}{c} \cdot \frac{L_1 - L}{2} \sin \bar{\varphi}\right\} \\ &\quad + \frac{L_2}{L} S(x, y; \theta_2) \\ &\quad \times \exp\left\{-j\frac{4\pi f_0}{c} \cdot \frac{L - L_2}{2} \sin \bar{\varphi}\right\}. \end{aligned} \quad (21)$$

Note that the “*translatability*” and “*additivity*” can be extended to the cases of more ASCs. If N ASCs have the same parameters except for the two position parameters, they have the same shape in the image domain. An ASC with length L can be a linear combination of two ASCs, each of which can be seen as a linear combination of two other ASCs. By repeating this process, an ASC with length L can be expressed as a linear combination of any number of ASCs.

B. Dictionary Reduction

Given Properties 1 and 2, there are opportunities to reduce the size of the dictionary Φ by removing atoms that can be synthesized using the other atoms in Φ . According to Property 1, an atom of the dictionary with the position parameters (x_q, y_q) can be obtained from the atom with the position parameter $(0, 0)$, as given by

$$\begin{aligned} D(x, y; (\sim, x_q, y_q, \sim)) \\ = D(x - x_q, y - y_q; (\sim, 0, 0, \sim)) e^{-j\frac{4\pi f_0}{c}x_q}. \end{aligned} \quad (22)$$

By substituting (22) into (8), we have

$$\begin{aligned} \iint R(x, y) \hat{D}^*(x, y; \theta_q) dx dy \\ = e^{j\frac{4\pi f_0}{c}x_q} \iint R(x, y) \hat{D}^*(x - x_q, y - y_q; \theta_{qo}) dx dy \\ = e^{j\frac{4\pi f_0}{c}x_q} (R \otimes \bar{\bar{D}})(x_q, y_q), \end{aligned} \quad (23)$$

where

$$\begin{aligned} \theta_q &= (A_q, \alpha_q, x_q, y_q, 2N_q\Delta L, \bar{\varphi}_q, \gamma_q); \\ \theta_{qo} &= (A_q, \alpha_q, 0, 0, 2N_q\Delta L, \bar{\varphi}_q, \gamma_q); \\ \bar{\bar{D}}(x, y; \theta_{qo}) &= D^*(-x, -y; \theta_{qo}). \end{aligned} \quad (24)$$

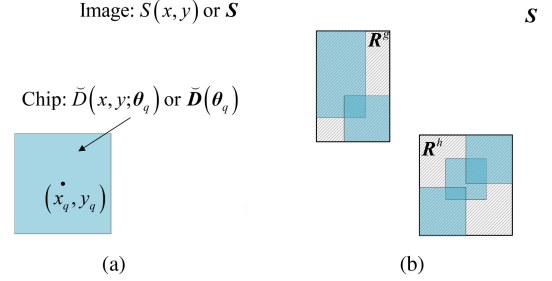


Fig. 1. Illustration on the construction of chips: (a) An example of a chip (marked in blue) in a SAR image; and (b) An example of two non-overlapping clusters of atoms, R^g and R^h , one of which contains two image chips (marked in blue) and the other contains three image chips. A cluster of atoms is the smallest rectangular region containing atoms which are all connected with overlapping areas, and it must not be further divided into smaller non-overlapping clusters.

It is indicated in (23) that the integral of the product of $R(x, y)$ and the dictionary atoms with the position parameter (x_q, y_q) can be obtained by the convolution of $R(x, y)$ and the flipped conjugate of the dictionary atoms with the position parameter $(0, 0)$. Hence, the dictionary atoms with the position parameters other than $(0, 0)$ can be suppressed from Φ .

According to Property 2, an atom with the length parameter $2N_q\Delta L$ can be obtained by a linear combination of the atoms with the length parameter ΔL , as given by

$$\begin{aligned} D(x, y; (\sim, 2N_q\Delta L, \bar{\varphi}_q, \sim)) &= \frac{1}{2N_q} \sum_{n=1}^{N_q} \\ &\times \left[e^{-j\frac{4\pi f_0}{c}\Delta x_n} D(x - \Delta x_n, y + \Delta y_n; (\sim, \Delta L, \bar{\varphi}_q, \sim)) \right. \\ &\quad \left. + e^{+j\frac{4\pi f_0}{c}\Delta x_n} D(x + \Delta x_n, y - \Delta y_n; (\sim, \Delta L, \bar{\varphi}_q, \sim)) \right], \end{aligned} \quad (25)$$

where $\Delta x_n = -(n - \frac{1}{2})\Delta L \sin(\bar{\varphi}_q)$, and $\Delta y_n = -(n - \frac{1}{2})\Delta L \cos(\bar{\varphi}_q)$. To this end, we only need to design the dictionary that only keeps the atoms with a length of ΔL .

In addition to the two new properties, another property of ASCs in the image domain (c.f. the frequency domain) is the good energy concentration of an ASC in a small area [33]. We can extract only a small segment, referred to as “chip,” to replace the atom in the dictionary, as shown in Fig. 1(a). The chip is large if the length parameter L of the dictionary element is large. The chip is small otherwise.

From (22), (25) and the high energy concentration in the image domain, we can replace $R(x, y)$ and $\hat{D}(x, y; \theta)$, respectively, with two chips extracted from them in (9). Accordingly, we can design a much more compact and efficient dictionary Φ_o (than Φ), as given by

$$\Phi_o = \left\{ \hat{\bar{D}}(x, y; \theta_q) \mid \theta_q \in \Theta_o \right\}, \quad (26)$$

where Θ_o is a subset of Θ . In particular, $\Theta_{L_{dis}} = \{\Delta L\}$, $\Theta_x = \{0\}$, and $\Theta_y = \{0\}$.

Here, Φ_o only has $(N_{\bar{\varphi}} + 1)$ atoms. Since the length parameters of the atoms in Φ_o are either 0 or ΔL , the atoms of Φ_o are all small chips. Φ_o requires a much smaller memory than Φ . For example, $N_{\bar{\varphi}} = 16$ for a typical 512×512 SAR image, while

the typical chips in Φ_o are 8×8 pixels. The total requirement of memory would be about 2 MB (as compared to 32 TB, as mentioned in Section III).

C. Acceleration of Atom Selection

As shown in (23), we can implement the inner product in (8) in the standard NOMP by using convolution. If the length parameter of θ_q is 0 or ΔL , then

$$C(x, y; \theta_q) = S \otimes \hat{D}^*(-x, -y; \theta_o), \hat{D}(x, y; \theta_o) \in \Phi_o. \quad (27)$$

If the length parameter of θ_q is $2N\Delta L$, then we can use Property 2 to prove that the convolution of the residual image $R(x, y)$ and the flipped conjugate of the atoms with the length of $2N_q\Delta L$ and the position of (0, 0) can be obtained by taking a linear combination of the convolution of the residual image and the flipped conjugate of the atoms with the length of ΔL in Φ_o . This is because convolution is a linear operator and can be implemented with a summation. $C(x, y; \theta_q)$ can be rewritten in a recurrence expression (28), shown at the bottom of this page, where $\|D_L\|_2 = \|D(x, y; \sim, L, \sim)\|_2$ for brevity. By using convolutions to implement the inner product in (8), the computation complexity of the atom selection can be reduced from the original $O(N_{\varphi}N_LN_xN_yMN)$ in (8) to $O(N_{\varphi}N_LMN)$.

D. Acceleration of Least-Square Estimation

As mentioned in Section III, the least-square estimation involves a pseudo-inversion; see (12). As illustrated in Fig. 1(b), the ASCs can exhibit strong locality in the image domain. The coefficients associated with the selected atoms in (11) account for the amplitudes of the atoms and are expected to minimize the square error of the reconstructed image. To this end, the least-square estimation of the coefficients can be decoupled losslessly between different, non-overlapping clusters of atoms (or

ASCs). The reason for the lossless decoupling of the coefficient estimation is that the coefficient of an atom can only affect the square error of the cluster containing the atom. Therefore, the coefficient has no impact on any other non-overlapping clusters.

By decoupling the least-square estimation between non-overlapping clusters of atoms, the pseudo-inversion required in (12) only needs to be carried out on a cluster basis. The complexity is only $O(i_c^2 M_c N_c)$ for an $M_c \times N_c$ -dimensional cluster containing i_c atoms. Generally, $i_c < i$ and $M_c N_c \ll MN$. Therefore, the complexity is substantially lower than the original $O(i^2 MN)$ in (12). Moreover, the reduced complexity $O(i_c^2 M_c N_c)$ is typically so small and comparatively negligible, as compared to those in the other steps (e.g., Steps 1 and 2 in Algorithm 1). As a result, the overall complexity of the proposed algorithm now tends to be dominated by Steps 1 and 2. It can become linear to the number of significant ASCs (rather than growing quadratically).

V. EXPERIMENT

In this section, we conduct extensive experiments on three types of data to validate the proposed algorithm, referred to as “PA” in all figures. First, we utilize synthetic backscatter data generated directly from the ASC model to demonstrate the validity, accuracy, and noise robustness of the algorithm. Second, we use the XPATCH backhoe data to test the algorithm. Third, we use a measured X-band SAR image chip of a T72 tank chosen from the Moving and Stationary Target Acquisition and Recognition (MSTAR) dataset [41], to test the practical performance of the proposed algorithm.

The experiments are run on a computer with Intel coreTM i5-4590 CPU@3.30 GHz, 8 G memory, and software MATLAB (R2017a). The existing image-domain (EID) algorithm developed in [26] is used for a comparison purpose.

$$\begin{aligned} C(x, y; (\sim, 2N_q\Delta L, \sim)) &= \iint R(u, v) \hat{D}^*(u - x, v - y; (\sim, 2N_q\Delta L, \sim)) dudv \\ &= \frac{1}{2N_q \|D_{2N_q\Delta L}\|_2} e^{+j\frac{4\pi f_0}{c}x} \iint R(u, v) \sum_{n=1}^{N_q} \|D_{\Delta L}\|_2 \left[e^{+j\frac{4\pi f_0}{c}\Delta x_n} \hat{D}^*(u - \Delta x_n - x, v + \Delta y_n - y; (\sim, \Delta L, \sim)) \right. \\ &\quad \left. + e^{-j\frac{4\pi f_0}{c}\Delta x_n} \hat{D}^*(u + \Delta x_n - x, v - \Delta y_n - y; (\sim, \Delta L, \sim)) \right] dudv \\ &= \frac{\|D_{\Delta L}\|_2}{2N_q \|D_{2N_q\Delta L}\|_2} e^{+j\frac{4\pi f_0}{c}x} \sum_{n=1}^{N_q} \left[e^{+j\frac{4\pi f_0}{c}\Delta x_n} C(x + \Delta x_n, y - \Delta y_n; (\sim, \Delta L, \sim)) \right. \\ &\quad \left. + e^{-j\frac{4\pi f_0}{c}\Delta x_n} C(x - \Delta x_n, y + \Delta y_n; (\sim, \Delta L, \sim)) \right] \\ &= \frac{1}{2N_q \|D_{2N_q\Delta L}\|_2} \left\{ 2(N_q - 1) \|D_{2(N_q-1)\Delta L}\|_2 C(x, y; (\sim, 2(N_q - 1)\Delta L, \sim)) \right. \\ &\quad + e^{+j\frac{4\pi f_0}{c}x} \|D_{\Delta L}\|_2 \left[e^{+j\frac{4\pi f_0}{c}\Delta x_{N_q}} C(x + \Delta x_{N_q}, y - \Delta y_{N_q}; (\sim, \Delta L, \sim)) + \right. \\ &\quad \left. \times e^{-j\frac{4\pi f_0}{c}\Delta x_n} C(x - \Delta x_{N_q}, y + \Delta y_{N_q}; (\sim, \Delta L, \sim)) \right] \left. \right\} \end{aligned} \quad (28)$$

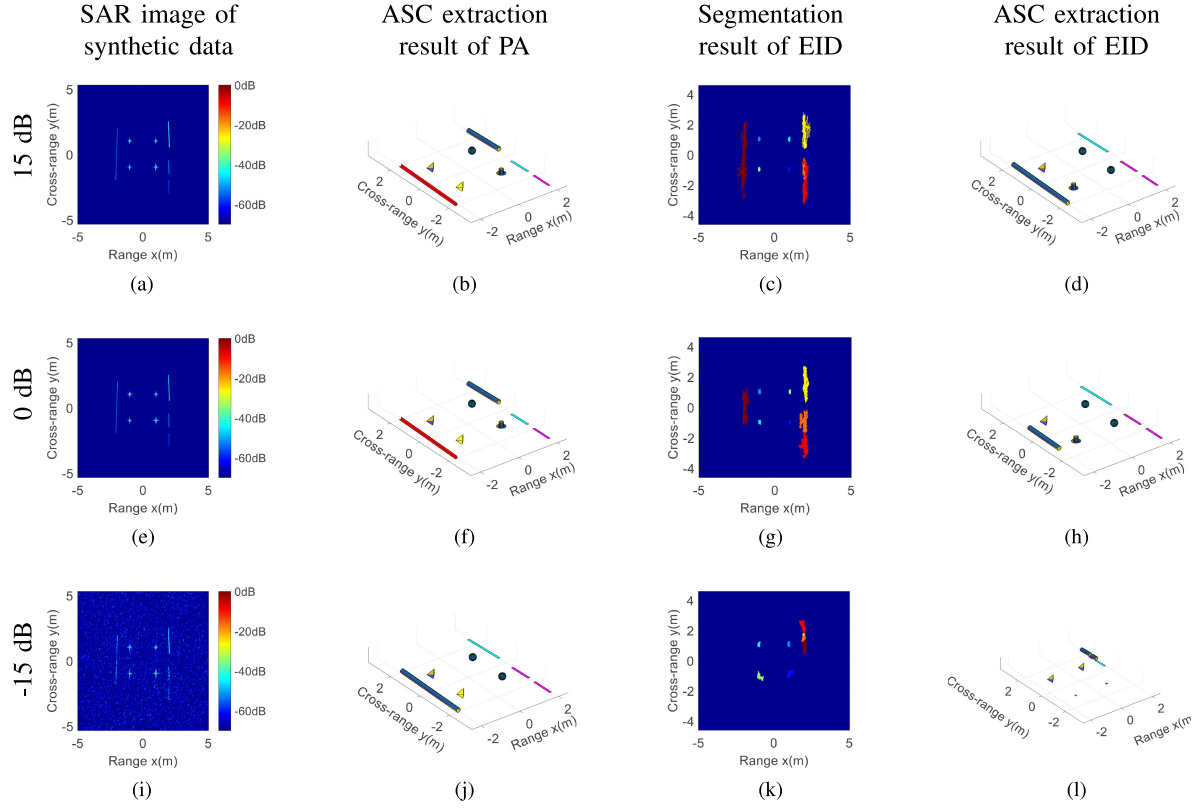


Fig. 2. Comparison between the proposed algorithm and EID [26]: (a), (e), and (i) are the SAR images; (b), (f), and (j) are the iconic representations of ASC extraction results by the proposed algorithm; (c), (g), and (k) are the segmentation results of the modified watershed algorithm [26]; and (c), (f), and (j) are the iconic representations of ASC extraction results by the EID algorithm [26].

TABLE II
SYSTEM CONFIGURATION OF SYNTHETIC BACKSCATTER DATA

Center Frequency	9.6 GHz
Bandwidth	3 GHz
Variation of Aspect Angle	$-5^\circ \sim 5^\circ$
Frequency Step	15 MHz
Aspect Angle Step	0.085°

A. Synthetic Backscatter

We synthesize the radar backscatter of 8 ASCs, including all the canonical scattering geometries, as shown in Table I. The simulated system configurations are collated in Table II, and the parameters of the ASCs are listed in Table III. We add a complex additive white Gaussian noise to the frequency domain of synthetic backscatter data with the SNR of 15 dB, 0 dB, and -15 dB. The images produced by the BP algorithm [40] are displayed in Fig. 2(a), (e), and (i). The proposed algorithm and the EID algorithm are used to extract the ASCs from the images.

The dictionary configuration of the proposed algorithm is $\Theta_{\bar{\varphi}} = \{N \cdot 0.5^\circ | N = -10, -9, \dots, 10\}$, and $\Delta L = 0.2$. The estimated parameters are listed in Table III. Fig. 2(b), (f), and (j) show the iconic representation of the results. We can see that all the parameters are accurately estimated when the SNR is 15 dB. The estimations of the amplitude of A , length L , position (x, y) ,

and orientation $\bar{\varphi}$ are robust to the noise. These parameters are accurately estimated even when the SNR is -15 dB. However, the estimations of the phases of A , α , and γ are susceptible to the noise, especially when the SNR is low.

Fig. 2 and Table III present the results of the EID algorithm. Fig. 2(c), (g), and (k) show the image segmentation results under different SNRs. The images are segmented by the modified watershed algorithm developed in [26]. It can be seen that the ASCs are successfully separated under the SNR of 15 dB. However, under the SNR of 0 dB, the first ASC is not correctly separated. Under the SNR of -15 dB, although the localized ASCs are correctly separated, the distributed ASCs are incorrectly separated. This is because the pixels of the distributed ASCs in the images have small amplitudes, as compared to the localized ASCs. They are likely to be corrupted by the noise. Table III lists the estimated parameters of the EID algorithm when the SNR is 15 dB and 0 dB. It can be found that the amplitude of A , length L , position (x, y) , and orientation $\bar{\varphi}$ are reasonably accurate; but the phases of A , α , and γ are not. In addition, the length of the first ASC is estimated to be 2.1078 m under 0 dB SNR, while the ground truth is 4 m. This is because only part of the ASC is separated in the image segmentation process, as can be seen from Fig. 2(g).

We run 100 independent tests for the proposed algorithm and the EID algorithm using the simulated ASCs, and evaluate the root mean square errors (RMSEs) of both algorithms. As shown

TABLE III
COMPARISON BETWEEN THE PROPOSED ALGORITHM AND THE STATE-OF-THE-ART EID [26] ON THE EXTRACTED RESULTS OF SYNTHETIC BACKSCATTER DATA

ASC No.	Algorithm	SNR	Detected Shape	Parameters						
				A	α	x	y	L	$\bar{\varphi}$	γ (1E-11)
1	PA	Ground Truth	Dihedral	4.4721 $\angle$ 0.4636	1	-2	0	4	-2	0
		15 dB	Dihedral	4.4664 $\angle$ 0.4436	1	-2.0001	-0.0005	3.9992	-1.9997	0
		0 dB	Dihedral	4.4593 $\angle$ 0.2319	1	-2	-0.0014	4.0099	-2.0008	0
		-15 dB	Cylinder	4.4434 $\angle$ 0.4528	0.5	-2.0028	-0.0437	3.9905	-2.0057	0
	EID	15 dB	Cylinder	4.4589 $\angle$ 2.0060	0.5	-2.0074	-0.0198	3.9491	-1.6166	0
		0 dB	Cylinder	2.4153 $\angle$ 1.7515	0.5	-2.0123	-0.1485	2.1078	-1.62	0
2	PA	Ground Truth	Cylinder	4 $\angle$ 0	0.5	2	1.5	2	2	0
		15 dB	Cylinder	4.0023 $\angle$ 0.0005	0.5	2	1.4996	1.9992	2.0000	0
		0 dB	Cylinder	4.0653 $\angle$ 0.1350	0.5	2.0001	1.5013	1.9952	2.0015	0
		-15 dB	EB	3.8542 $\angle$ 0.2830	0	1.9994	1.4954	2.0012	1.9893	0
	EID	15 dB	EB	4.2126 $\angle$ -2.2309	0	1.9825	1.4619	1.977	1.8698	0
		0 dB	EB	4.2040 $\angle$ -2.3076	0	1.9825	1.4595	1.9725	1.8701	0
3	PA	Ground Truth	EB	1.4142 $\angle$ 0.7854	0	2	-1	1.1	1.3	0
		15 dB	EB	1.4176 $\angle$ 0.9357	0	2.0004	-0.9998	1.099	1.3051	0
		0 dB	EB	1.4060 $\angle$ 1.2074	0	2.0003	-1.0061	1.0891	1.3093	0
		-15 dB	ED	1.2911 $\angle$ 0.7794	-0.5	1.9983	-1.004	1.0988	1.3169	0
	EID	15 dB	ED	1.5429 $\angle$ 2.3332	-0.5	1.9827	-1.0073	1.0863	1.2572	0
		0 dB	ED	1.5046 $\angle$ 2.4724	-0.5	1.9831	-1.0085	1.0857	1.25	0
4	PA	Ground Truth	ED	1 $\angle$ 0	-0.5	2	-2.5	1.05	0	0
		15 dB	ED	0.9972 $\angle$ 0.0454	-0.5	2.0001	-2.501	1.0469	0	0
		0 dB	ED	1.0044 $\angle$ 0.4230	-0.5	2.0001	-2.4992	1.0592	0	0
		-15 dB	ED	1.1227 $\angle$ -1.6006	-0.5	1.9962	-2.4998	1.0803	0.0268	0
	EID	15 dB	ED	1.1066 $\angle$ -0.7438	-0.5	1.9827	-2.4877	1.0377	0.1289	0
		0 dB	ED	1.1480 $\angle$ -0.7301	-0.5	1.9828	-2.4892	1.0318	0.1228	0
5	PA	Ground Truth	Trihedral	1 $\angle$ 0	1	-1	-1	0	0	0
		15 dB	Trihedral	0.9945 $\angle$ -0.0636	1	-1	-0.9999	0	0	0.0059
		0 dB	Trihedral	0.9960 $\angle$ -0.2483	1	-1.0003	-0.9999	0	0	0.1301
		-15 dB	Trihedral	0.9696 $\angle$ -0.6667	1	-1.0007	-1.0034	0	0	0.6163
	EID	15 dB	Top Hat	1.0199 $\angle$ 0.7731	0.5	-1.0099	-1.0067	0	0	-1.6268
		0 dB	Top Hat	1.0239 $\angle$ 0.7817	0.5	-1.0099	-1.0059	0	0	-2.02
6	PA	Ground Truth	Top Hat	1.4142 $\angle$ -0.7854	0.5	1	-1	0	0	0
		15 dB	Top Hat	1.4068 $\angle$ -0.7411	0.5	1.0001	-0.9999	0	0	0.0176
		0 dB	Top Hat	1.4110 $\angle$ -0.7518	0.5	1	-1	0	0	-0.1813
		-15 dB	Sphere	1.4034 $\angle$ -0.5599	0	0.9995	-0.9993	0	0	-0.5452
	EID	15 dB	Sphere	1.4406 $\angle$ 0.4665	0	0.9851	-1.0067	0	0	-1.6844
		0 dB	Sphere	1.4407 $\angle$ 0.4489	0	0.985	-1.0064	0	0	-1.5927
7	PA	Ground Truth	Sphere	1 $\angle$ 0	0	1	1	0	0	2
		15 dB	Sphere	0.9954 $\angle$ 0.1275	0	1	1.0001	0	0	1.9715
		0 dB	Sphere	1.0024 $\angle$ -0.0555	0	0.9999	1.0003	0	0	1.8404
		-15 dB	Sphere	1.0192 $\angle$ 1.0277	0	1.0019	1.002	0	0	2.2169
	EID	15 dB	Sphere	1.0218 $\angle$ 2.4971	0	0.985	0.968	0	0	-0.0038
		0 dB	Sphere	1.022 $\angle$ 2.4263	0	0.9848	0.9678	0	0	-0.3997
8	PA	Ground Truth	CD	1 $\angle$ 0	-1	-1	1	0	0	1
		15 dB	CD	0.9949 $\angle$ 0.1593	-1	-0.9999	1.0001	0	0	1.0268
		0 dB	CD	0.9936 $\angle$ 0.1092	-1	-1.0001	1.0004	0	0	1.0852
		-15 dB	CD	0.9696 $\angle$ -0.6667	-1	-1.0007	-1.0034	0	0	0.6163
	EID	15 dB	CD	1.0300 $\angle$ 2.0021	-1	-1.01	0.9681	0	0	-0.7788
		0 dB	CD	1.0385 $\angle$ 1.9343	-1	-1.0101	0.9679	0	0	-0.713

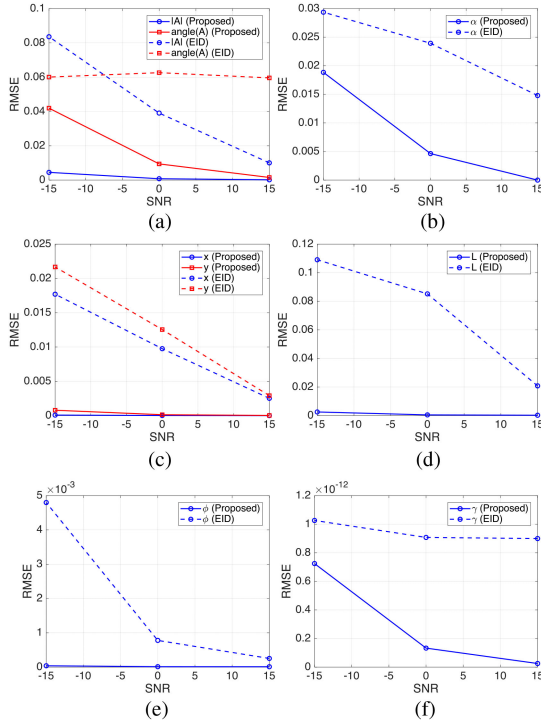


Fig. 3. The root mean square error (RMSE) between the real values of the simulated ASC and the values estimated by different algorithms.

TABLE IV
BACKHOE DATA SYSTEM CONFIGURATION

Center Frequency	10 GHz
Bandwidth	5.9 GHz
Variation of Aspect Angle	$-10^\circ \sim 10^\circ$
Frequency Step	11.557 MHz
Aspect Angle Step	0.0714°

in Fig. 3, the RMSEs of the proposed algorithm are much smaller than those of the EID algorithm under all tested parameters. By comparing the results of two algorithms, it is found that the proposed algorithm is more robust to the noises than the EID algorithm. The EID algorithm fails to correctly estimate the frequency dependence parameter α , even when the SNR is high, e.g., 15 dB.

B. XPATCH Backhoe Results

We proceed with experiments on a publicly released dataset, Backhoe Data Version 1.0 [42]. Fig. 4(a) is the CAD model of the dataset. The system configurations are provided in Table IV. The SAR image produced by the BP algorithm is displayed in Fig. 4(c). The dictionary configuration of the proposed algorithm is $\Theta_{\bar{\varphi}} = \{N \cdot 0.2^\circ | N = -50, -49, \dots, 50\}$ and $\Delta L = 0.1$. The dictionary only requires 0.12 MB memory. Fig. 5 shows the ASC extraction results of both the proposed algorithm and the EID algorithm.

Fig. 5(e) and (f) show the CPU time and the residual ratio with a growing number of iterations in the two algorithms. Table VII compares the CPU time between the proposed algorithm, the

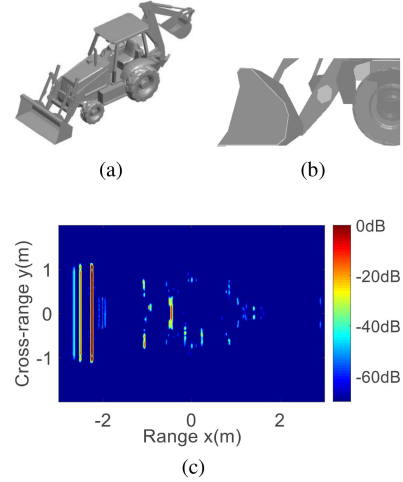


Fig. 4. Illustration on a SAR image: (a) the CAD model of backhoe; (b) The side view of the front shovel of the CAD model; and (c) the SAR image produced by the BP algorithm [40].

TABLE V
ESTIMATED PARAMETERS FOR THE FRONT SHOVEL

	Parameters					
	$ A $	α	x	y	L	$\bar{\varphi}$
PA	5.7757	1	-2.643	0.0119	2.0543	0
	16.8492	0	-2.5018	0.0019	1.9829	0
	36.8299	-0.5	-2.2413	0.0008	1.9809	0
EID	7.3852	0.5	-2.6402	-0.0087	2.0608	0.0043°
	21.1545	-0.5	-2.5001	-0.0033	1.9224	0.0043°
	42.0442	-0.5	-2.2405	0.0164	1.9553	0.0043°

TABLE VI
ESTIMATED PARAMETERS FOR THE BACKHOE

	Parameters					
	$ A $	α	x	y	L	$\bar{\varphi}$
PA	0.5738	-0.5	2.9213	-0.0006	0.8303	0
EID	11.7312	-0.5	2.9045	-0.2565	0.2645	0.0148°
	11.7697	-0.5	2.9048	0.2388	0.268	0

TABLE VII
COMPARISON OF THE PROPOSED ALGORITHM WITH THE STATE-OF-THE-ART IMAGE-DOMAIN EID ALGORITHM [26] AND THE STATE-OF-THE-ART FREQUENCY-DOMAIN ISBL ALGORITHM [32]

Algorithm	CPU time grow	Time cost in 40 iterations
PA	linearly	8 min
ISBL [31]	quadratically [31, Fig. 16]	900 min [31, Fig. 16]
EID	linearly	2.56 min

EID algorithm, and the ISBL algorithm [32]. It is shown that the CPU times of the proposed algorithm and the EID algorithm grow linearly with the number of iterations, while the CPU time of the ISBL algorithm grows quadratically. The runtime of the proposed algorithm is only 8 min for 40 iterations, while the runtime of the ISBL algorithm is 900 min. The proposed algorithm is efficient due to the acceleration techniques proposed in Section VI.

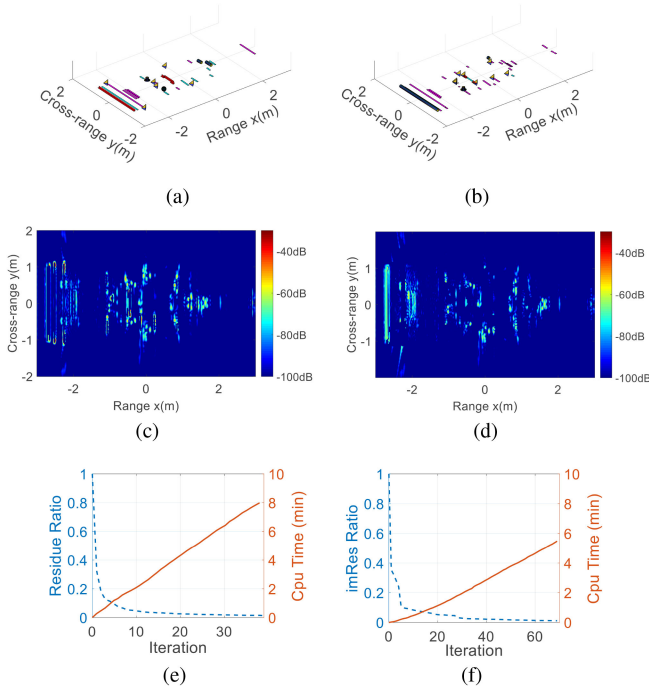


Fig. 5. Comparison between the proposed algorithm (left) and the EID [26] (right) based on the Backhoe data: (a) and (b) are the iconic representations of the ASC extraction results; (c) and (d) are the residual images; and (e) and (f) provide the CPU time and the residual ratio (i.e., the ratio of the energy between the residual image and the original SAR image) with the increasing number of iterations.

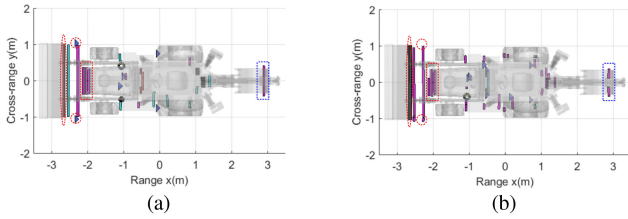


Fig. 6. Projection of the ASC models extracted from: (a) the proposed algorithm and (b) the state-of-the-art EID algorithm [26] on the CAD model (top view).

It is noticed that the EID algorithm takes a shorter time than the proposed algorithm during each iteration, although the CPU time of both algorithms grows linearly with iterations. The most time-consuming part of the proposed algorithm is the refinement in Step 2 in Section III because we use the BP algorithm to implement the imaging operator $\mathcal{B}\{\cdot\}$. The BP algorithm is generic and known to generate quite accurate SAR images. The efficiency of the proposed algorithm can be further improved if a more efficient image generation algorithm, other than BP, is applied.

We also plot the CAD model and the extracted ASCs in Fig. 6. It can be seen that there are three ASCs with a length of about 2 m. But only one is in the front shovel, and the other two are outside the shovel. The reason is that the front shovel has a relatively complex shape, as can be observed in Fig. 4(b). Some electromagnetic waves are reflected multiple times by the front shovel, causing the two ASCs to deviate from the front shovel.

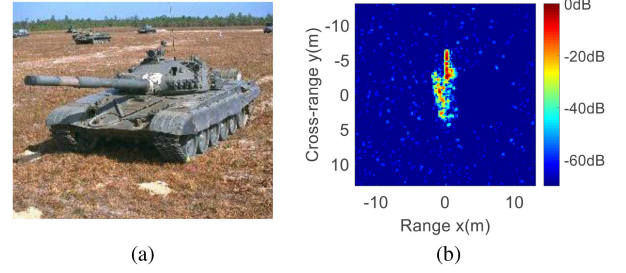


Fig. 7. Measured data of a T72 tank: (a) The Photo of T72; (b) the SAR image of T72.

TABLE VIII
MSTAR DATA SYSTEM CONFIGURATION

Center Frequency	9.6 GHz
Bandwidth	0.591 GHz
Variation of Aspect Angle	$-1.76^\circ \sim 1.76^\circ$
Frequency Step	4.6535 MHz
Aspect Angle Step	0.0278°

The ASC marked with the red ellipse is dihedral in Fig. 6(a) and cylindrical in Fig. 6(b). The ground truth is dihedral because, as observed in Fig. 4(b), the front shovel has dihedral structures, not cylinder structures. It is also found that the front shovel has corner structures whose corresponding ASCs are successfully extracted by the proposed algorithm, as shown in the red circles in Fig. 6(a). In contrast, the two ASCs are not extracted by the EID algorithm. The proposed algorithm outperforms the EID algorithm in terms of ASC extraction accuracy.

For a quantitative verification, we compare the estimated parameters of the front shovel by running the proposed algorithm and the EID algorithm in Table V, and the estimated parameters of the backhoe in Table VI. The length of the backhoe is estimated to be 0.7615 m based on the positions and lengths of the two ASCs extracted by the EID algorithm, as shown in Table VI. The actual length of the front shovel is 2 m. The actual length of the backhoe is 0.75 m. The front shovel lengths estimated by the two algorithms are very close to the actual length. The backhoe length estimated by the EID algorithm is closer to the ground truth than the proposed algorithm. The reason is that the two separate ASCs are misjudged to a single ASC by the proposed algorithm.

C. MSTAR Target

The proposed algorithm is tested using a measured X-band SAR image chip of a T72 tank. The image is chosen from the MSTAR dataset [43] and the barrel of the tank is broadside to the radar. The corresponding optical image is shown in Fig. 7(a) and the SAR image is provided in Fig. 7(b). The system configurations are listed in Table VIII.

The proposed algorithm and the EID algorithm are used to extract the ASCs from the MSTAR data. The dictionary configuration of the proposed algorithm is $\Theta_{\varphi} = \{N \cdot 0.1^\circ | N = -17, -16, \dots, 17\}$ and $\Delta L = 0.1$. The dictionary only requires 0.03 MB memory. Fig. 8 shows the ASC extraction results of

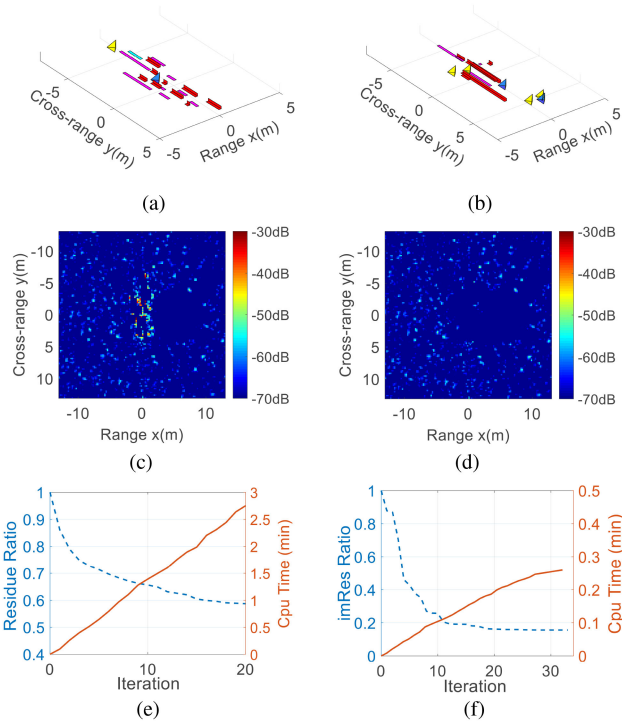


Fig. 8. Comparison between the proposed algorithm (left) and EID (right) based on MSTAR data: (a) and (b) are the iconic representations of the ASC extraction results; (c) and (d) are the residual images; and (e) and (f) provide the CPU time and the residual ratio with the increasing number of iterations.

TABLE IX
FINAL PARAMETER ESTIMATES FOR THE EXERTED BARREL

	Parameters					
	$ A $	α	x	y	L	$\bar{\varphi}$
PA	0.5475	-0.5	0.1019	-4.1969	3.2	0.3839°
EID	0.5873	-0.5	0.1239	-4.2946	3.0092	-0.0079°

the proposed algorithm and the EID algorithm. Fig. 8(e) and (f) provide the CPU time and residual ratio with the growing number of iterations in both algorithms. By comparing Fig. 8(a) and (b), we can see the result of the proposed algorithm is closer to the ground truth, since T72 has many distributed structures, as shown in Fig. 7(a). The proposed algorithm extracts more distributed ASCs than the EID algorithm. As shown in Fig. 8(e) and (f), the CPU times of the proposed algorithm and the EID algorithm grow linearly with the number of iterations. The proposed algorithm is as efficient as the EID algorithm.

Table IX lists the parameters of the ASC of the exerted barrel. The actual length of the exerted barrel is about 5.4 m. It is seen that the lengths estimated by the two algorithms are both inaccurate. The reason can be due to the fact the barrel is not an ideal cylinder, as shown in Fig. 7(a). The α value of the ASC corresponding to the exerted barrel is expected to be 0.5 since part of the exerted barrel can still be seen as a cylinder. However, both the proposed algorithm and the EID algorithm give inaccurate α values. This is due to the effect of clutter and the finite radar bandwidth. As discussed in Section V-A, α is

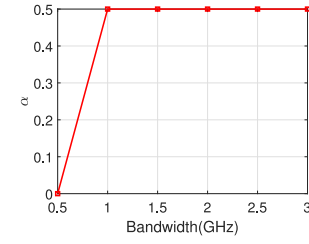


Fig. 9. The α value estimated by the proposed algorithm under different bandwidths.

sensitive to the noise and can be inaccurately estimated under a low SNR. To illustrate the effect of the radar bandwidth on the estimation of α , we synthesize the radar backscatter of an ASC with the following parameters:

$$A = 4, \alpha = 0.5, x = 0, y = 0, L = 4, \bar{\varphi} = 0. \quad (29)$$

The configuration of the radar system is as listed in Table II, except for the bandwidth. The SNR is set to 15 dB. We synthesize the radar backscatter under different bandwidths and use the proposed algorithm to estimate the α value. The estimation results are shown in Fig. 9, where the proposed algorithm is increasingly accurate with the growing bandwidth.

VI. CONCLUSION

In this paper, a novel ASC extraction algorithm is proposed by exploiting the sparse representation of SAR images in the image domain at a pixel level. On the one hand, the proposed algorithm is superior to existing frequency-domain algorithms in terms of accuracy and noise resistance. On the other hand, our algorithm inherits a low memory requirement and moderate computation complexity from state-of-the-art image-domain algorithms. Experimental results show that the proposed algorithm is more robust and accurate than existing image-domain algorithms and more computationally efficient than existing frequency-domain algorithms.

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